

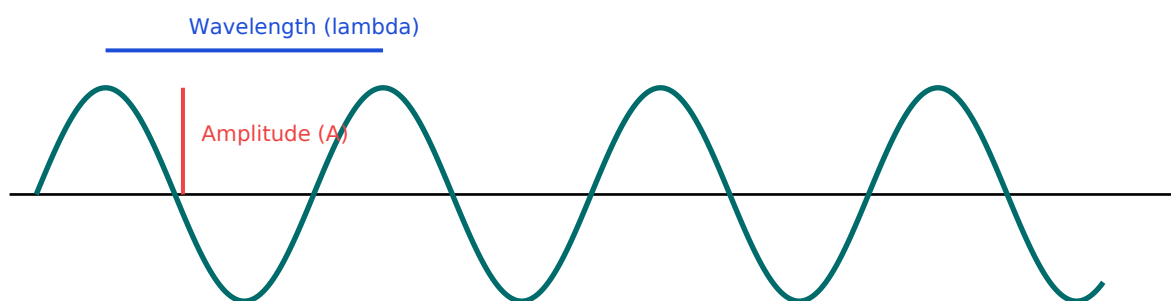
JDScience BTEC Level 3 Waves in Communication Worksheet

Unit 1C - wave properties, wave equation, phase, interference and diffraction.

Learning aims

- Compare transverse and longitudinal waves.
- Use wave quantities: amplitude, frequency, period, wavelength and wave speed.
- Apply $v = f \lambda$ in calculations.
- Explain phase difference, coherence, superposition and interference.
- Describe diffraction, diffraction gratings and electromagnetic waves in communication.

Wave basics



Transverse wave: displacement is perpendicular to direction of travel

Waves transfer energy from one place to another without transferring matter. In transverse waves, oscillations are perpendicular to the direction of energy transfer. In longitudinal waves, oscillations are parallel to the direction of energy transfer and include compressions and rarefactions.

Quantity	Meaning	Unit
Amplitude A	Maximum displacement from equilibrium	m
Frequency f	Number of waves per second	Hz
Period T	Time for one complete wave	s
Wavelength lambda	Distance between matching points on adjacent waves	m
Wave speed v	Distance travelled by wave per second	m/s

Wave equation and calculations

All waves obey $v = f \times \lambda$. Electromagnetic waves travel at approximately 3.0×10^8 m/s in a vacuum. Convert units carefully: kHz = $\times 1000$ Hz, MHz = $\times 1,000,000$ Hz, cm = $/100$ m, km = $\times 1000$ m.

Phase, coherence and interference

Phase difference measures how far one wave lags behind another, often in degrees or radians. Coherent waves have a constant phase difference. Superposition occurs when waves pass through one another and their displacements add.

Constructive interference occurs when coherent waves arrive in phase. Destructive interference occurs when coherent waves arrive out of phase.

Diffraction and diffraction gratings

Diffraction is the spreading of waves when they pass through a gap or around an obstacle. Maximum diffraction occurs when the gap is approximately the same size as the wavelength.

A diffraction grating contains many parallel slits and produces sharp bright maxima. The central maximum is the zero order. Higher orders occur when neighbouring waves have path differences of whole numbers of wavelengths.

Student worksheet tasks

A. Definitions

Define oscillation, compression, rarefaction, amplitude, frequency, period and wavelength.

B. Calculations

Calculate: 1) v for $f=10$ kHz and $\lambda=2$ m. 2) v for $f=4$ kHz and $\lambda=50$ cm. 3) f for $v=500$ m/s and $\lambda=25$ m. 4) λ for $v=11$ km/s and $f=5.5$ kHz.

C. Explanation

Explain why two people speaking at different pitch from the same side of a hall are heard at the same time.

D. Interference

Compare constructive and destructive interference and link each to path difference.

E. Diffraction grating

Explain why the central maximum is bright and why coloured/bright bands are produced at angles.